

- Requirements
 - Program must be able to allow the user to input a statement
 - Program must take the statement and count the number of specific type of coins and keep track of the obtained data
 - Program must account for base coin type and their plural forms
 - Based on the number of coins and their coin types, the program must calculate the total dollar amount

- Design(Pseudocode)

```
PRINT "Please enter your statement:"
```

```
READ user_input
```

```
SPLIT user_input INTO words
```

```
FOR EACH word IN words DO
```

```
  IF word is a number THEN
```

```
    SET count TO integer value of word
```

```
    CONTINUE to next word
```

```
  ENDIF
```

```
  IF word == "penny" OR word == "pennies" THEN
```

```
    SET cost TO 0.01
```

```
  ENDIF
```

```
  IF word == "nickel" OR word == "nickels" THEN
```

```
    SET cost TO 0.05
```

```
  ENDIF
```

```
  IF word == "dime" OR word == "dimes" THEN
```

```
    SET cost TO 0.10
```

```
  ENDIF
```

```
  IF word == "quarter" OR word == "quarters" THEN
```

```
    SET cost TO 0.25
```

```
  ENDIF
```

```
  ADD (count * cost) TO amount
```

```
  SET count TO 0
```

```
END FOR
```

```
PRINT amount
```

END

- Implementation

- Source code can be found in Final_Sprint_1.py

- Testing

Test Cases:

- 1 penny and 2 nickels | Result: 0.11
- 1 PENNY AND 2 NICKELS | Result: 0
- 4 dimes and 7 quarters | Result: 2.15
- 4 DIMES AND 7 QUARTERS | Result: 0
- 1 quarter and 3 pennies | Result: 0.28
- 1 QUARTER AND 3 PENNIES | Result: 0
- 21 pennies and 17 dimes and 52 quarters | Result: 14.91
- 21 PENNIES AND 17 DIMES AND 52 QUARTERS | Result: 0
- 95 dimes and 73 quarters and 22 nickels and 36 pennies | Result: 29.21
- 95 DIMES AND 73 QUARTERS AND 22 NICKELS AND 36 PENNIES | Result: 0
- 1 nickel and 17 quarters | Result: 4.3
- 1 NICKEL AND 17 QUARTERS | Result: 0
- 21 nickels and 15 pennies | Result: 1.2
- 21 NICKELS AND 15 PENNIES | Result: 0
- 1 dime and 1 nickel and 1 penny and 1 quarter | Result: 0.41000000000000003
- 1 DIME AND 1 NICKEL AND 1 PENNY AND 1 QUARTER | Result: 0
- 1 dime , 1 nickel . 1 penny , 1 quarter, Result: 0.41000000000000003

The logic does work for the most part, however there are a few issues. The most obvious is that if any capital letter is present, the program will fail to correctly compute the dollar amount. Secondly the format for some of the amounts returned by the program are not correct. For example some outputs are unusually long while others are cut one digit too short.